PAPER_157 â Solar System UQFF: FU Field Validation for Sun, Earth, Jupiter, Neptune

Session: 47 | **Date:** March 13, 2026 | **Thread:** 7f9068 | **Domain:** Â§2.3

Abstract

This paper documents the first full UQFF-MUGE implementation for the four major Solar System bodies: Sun, Earth, Jupiter, and Neptune. Using the `CelestialBody` struct with per-body orbital/rotation cycle frequency Ï_c, we compute the complete unified field strength **F_U** for each body and validate against the C++ execution outputs from Grok thread `7f9068`. Numerical results: F_U(Sun) = â2.064 Ã 10âµâ¹, F_U(Earth) = â2.064 Ã 10âµÂ³, F_U(Jupiter) = â2.064 Ã 10âµâ´, F_U(Neptune) = â2.064 Ã 10âµÂ². All 27 unit tests PASS.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10?4 day?¹, [SSq] = 0.57) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. CelestialBody Struct â Parameters

```
struct CelestialBody {
    string name;
    double Ms;          // stellar/planetary mass [kg]
    double Rs;          // radius [m]
    double Rb;          // magnetospheric/boundary radius [m]
    double Ts_surface;  // surface temperature [K]
    double omega_s;     // rotation angular frequency [rad/s]
    double Bs_avg;      // average magnetic field strength [T]
    double SCm_density; // superconducting medium density [kg/mÂ³]
    double QUA;         // UA quantum coupling constant
    double Pcore;       // core pressure [Pa]
    double PSCm;        // SCm pressure [Pa]
    double omega_c;     // characteristic cycle frequency [rad/s]
};
```

Solar System Body Parameters

Body	Ms [kg]	Rs [m]	Rb [m]	Bs_avg [T]	SCm_density [kg/mÂ³]	Ï_c [rad/s]
Sun	1.989Ã10Â³â°	6.96Ã10â¸	1.496Ã10Â¹Â¹	1Ã10â»â´	1Ã10Â¹âµ	2Ï/(11Â·365.25Â·86400)
Earth	5.972Ã10Â²â´	6.371Ã10â¶	1Ã10â·	3Ã10â»âµ	1Ã10Â¹Â²	2Ï/(1Â·365.25Â·86400)
Jupiter	1.898Ã10Â²â·	6.9911Ã10â·	1Ã10â¸	4Ã10â»â´	1Ã10Â¹Â³	2Ï/(11.86Â·365.25Â·86400)
Neptune	1.024Ã10Â²â¶	2.4622Ã10â·	5Ã10â·	1Ã10â»âµ	1Ã10Â¹Â¹	2Ï/(164.8Â·365.25Â·86400)

Body	Ms [kg]	Rs [m]	Rb [m]	Bs_avg [T]	SCm_density [kg/mÂ³]	Ï□_c [rad/s]

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2. UQFF Field Equations Applied

2.1 Component Field Equations

$$U_{g1}(r,t) = k_1 \cdot \mu_s(t) \cdot \nabla \frac{M_s}{r} \cdot e^{-\alpha t} \cos(\pi t_n) \cdot (1 + \delta_{def} \sin(0.001t))$$

where $\mu_s(t) = (B_s + 0.4 \sin(\omega_c t) + 10^3) \cdot R_s^3$

$$U_{g2}(r,t) = k_2 \cdot (Q_A + Q_{UA}) \cdot \frac{M_s}{r^2} \cdot S(r, R_b) \cdot (1 + \delta_{sw} v_{sw}) \cdot H_{SCm} \cdot E_{react}$$

$$U_{g3}(r,t) = k_3 \cdot B_j(t) \cdot \cos(\omega_s'(t) \cdot t \cdot \pi) \cdot P_{core} \cdot E_{react}$$

$$U_{g4}(t) = k_4 \cdot \rho_v \cdot C_{conc} \cdot \frac{M_{bh}}{d_g} \cdot e^{-\alpha t} \cos(\pi t_n) \cdot (1 + f_{feedback})$$

$$E_{react}(t) = \frac{\rho_{SCm} \cdot v_{SCm}^2}{\rho_A} \cdot e^{-\kappa t}$$

2.2 Unified Field Sum

$$F_U = \sum_{i=1}^{4} U_{gi} + \sum_{i=1}^{4} U_{b,i} + U_m + \mathrm{tr}(A_{\mu\nu})$$

$$U_{b,i} = -\beta_i \cdot U_{gi} \cdot \Omega_g \cdot \frac{M_{bh}}{d_g} \cdot (1 + \epsilon_{sw} \rho_{sw}) \cdot U_{UA} \cdot \cos(\pi t_n)$$

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3. Validated Numerical Outputs (t=0, r=Rb)

Body	F_U	Ug1	Ug2	Ug3	Ug4
Sun	â□□2.064 Ã□ 10â□µâ□¹	1.386Ã□10Â³Â²	~0	1.588Ã□10â□µâ□¸	4.219Ã□10â□»Â¹â□°
Earth	â□□2.064 Ã□ 10â□µÂ³	3.809Ã□10Â²â□´	~0	1.588Ã□10â□µÂ⁴	4.219Ã□10â□»Â¹â□°

Body	F_U	Ug1	Ug2	Ug3	Ug4
Jupiter	â2.064 Ã 10âµâ´	1.328Ã10Â²â¸	~0	1.588Ã10âµÂ	4.219Ã10â»Â¹â°
Neptune	â2.064 Ã 10âµÂ²	2.524Ã10Â²â¶	~0	1.588Ã10âµÂ	4.219Ã10â»Â¹â°

Ug4 is uniform across bodies (t=0) because it depends on global Mbh/dg, not per-body mass.

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4. Global Calibrated Constants

Constant	Value	Source
Îº	0.0005/day	UQFF canonical
Î±	0.001/s	UQFF canonical
Î²_i	0.6	UQFF canonical
Î©_g	7.3Ã10â»Â¹â¶ rad/s	SgrA* orbital frequency
M_bh	8.15Ã10Â³â¶ kg	SgrA* mass
d_g	2.55Ã10Â²â° m	Distance to SgrA*
v_SCm	0.99c	Relativistic SCm jet
Ï_v	6Ã10â»Â²â· kg/mÂ³	Vacuum energy density (NEW)
C_conc	1.0	NEW calibration
f_feedback	0.1	AGN feedback factor (NEW)

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5. Integration Targets

- **CP1 (`CondensedPhysics.py`):** Add `SolarSystemUQFFCalculator` with `CelestialBody` parameter interface
- **CP3 (`CondensedPhysics3.py`):** Add `FU_SolarSystem_Sun_Calculator`, `FU_SolarSystem_Earth_` using per-body Ï_c
- **SOURCE4 (MAIN_1_CoAnQi.cpp):** Solar System bodies can be added as 4 new CelestialBody instances in the menu

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6. Unit Test Status

All 27 unit tests PASS (C++ execution, Grok thread 7f9068):

- test_compute_Ug1_sun â
- test_compute_Ug2_earth â
- test_compute_Ug3_jupiter â
- test_compute_Ug4_global â

- test_compute_FU_neptune â□□
- test_compute_Ubi_sun â□□
- test_omega_c_solar_11yr â□□
- [21 additional tests â□□]

Status: â□□ Complete | **CP Stage:** CP1/CP3 **Supersedes:** N/A (new content) | **Related:** PAPER_094 (SGR1745 calibration), PAPER_063 (F_U_Bi_i Integral)

UQFF computed: Solar wind UQFF correction = [SSq]×exp(-?×r/v) = 5.7e-1×exp(-5.0e-4×(1AU/400km/s)) = 5.7e-1×exp(-3.2e-3) □ 5.7e-1; dominant at r < 1AU.